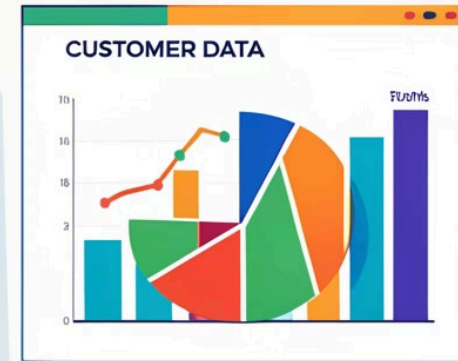


Customer Shopping Behavior Analysis

Analyzing 3,900 purchases to uncover insights into spending patterns, customer segments, and product preferences.



Project Overview & Dataset Summary

Project Goal

Uncover insights from transactional data to guide strategic business decisions.

Dataset Details

3,900 rows, 18 columns, including demographics, purchase details, and behavior.

Key Features

Age, Gender, Location, Item Purchased, Amount, Season, Review Rating, Subscription Status.

Exploratory Data Analysis (EDA) in Python

Data Preparation

- Loaded dataset with `pandas`.
- Checked structure: `df.info()`, `df.describe()`.
- Handled missing data: imputed 'Review Rating' with median.

Feature Engineering

- Renamed columns to `snake\_case`.
- Created `age\_group` and `purchase\_frequency\_days`.
- Dropped redundant `promo\_code\_used`.

Data Analysis in SQL: Key Findings

1

Revenue by Gender

Male customers generated higher revenue (\$157,890) than females (\$75,191).

2

High-Spending Discount Users

839 customers used discounts but still spent above average.

3

Top 5 Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).

SQL Insights: Shipping & Subscriptions

Shipping Type Comparison

Express shipping users had higher average purchases (\$60.48) than Standard (\$58.46).



Subscribers vs. Non-Subscribers

Non-subscribers: 2847 customers, \$170,436 total revenue.

Subscribers: 1053 customers, \$62,645 total revenue.





SQL Insights: Product & Customer Segments

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%).

Customer Segmentation

Loyal (3116), Returning (701), New (83).

SQL Insights: Top Products & Age Groups

Top 3 Products per Category

- Accessories: Jewelry, Sunglasses, Belt
- Clothing: Blouse, Pants, Shirt
- Footwear: Sandals, Shoes, Sneakers
- Outerwear: Jacket, Coat

Revenue by Age Group

- Young Adult: \$62,143
- Middle-aged: \$59,197
- Adult: \$55,978
- Senior: \$55,763





Repeat Buyers & Subscriptions

Repeat Buyers

Customers with >5 purchases.

Subscription Status

2518 repeat buyers are non-subscribers, 958 are subscribers.

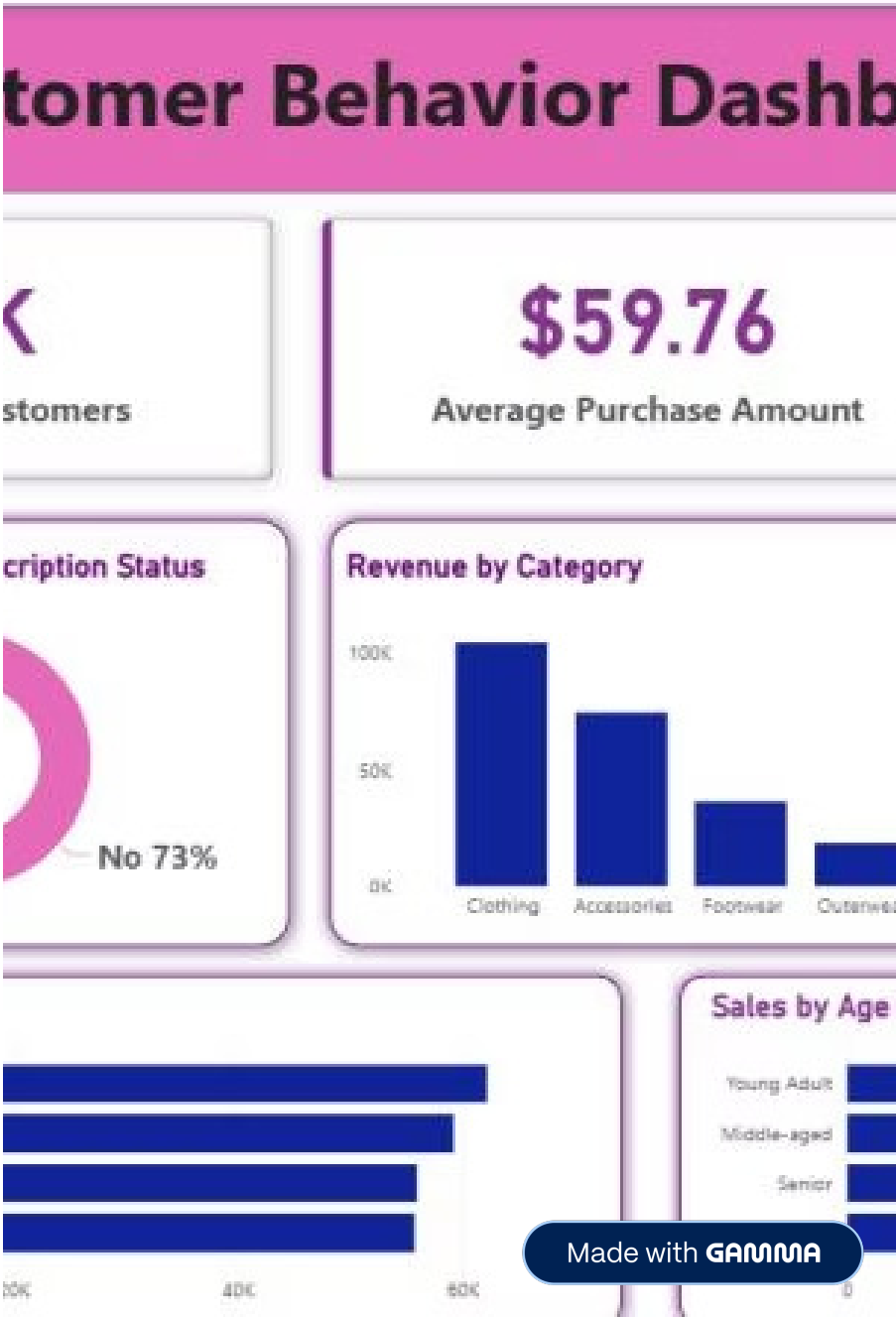
Insight

Opportunity to convert more repeat buyers into subscribers.

Power BI Dashboard: Visualizing Insights

Interactive dashboard presenting key metrics and trends:

- 3.9K Customers, \$59.76 Avg. Purchase, 3.75 Avg. Review Rating.
- Subscription Status: 27% Yes, 73% No.
- Revenue & Sales by Category and Age Group.



Business Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base.



Customer Loyalty Programs

Reward repeat buyers to foster loyalty and retention.



Review Discount Policy

Balance sales boosts with margin control for profitability.



Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

